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DS 120 Fall'25

Time: 50 minutes

Midterm 1
50 Points

1. Use truth table to determine whether the following two statements are logically equivalent: $(P \wedge Q) \Rightarrow R$ and $P \Rightarrow (Q \Rightarrow R)$. 6 points

Write your final conclusion as a statement.

P	Q	R	$P \wedge Q$	$(P \wedge Q) \Rightarrow R$	$Q \Rightarrow R$	$P \Rightarrow (Q \Rightarrow R)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	F	T	T	T
F	T	F	F	T	F	T
F	F	T	F	T	T	T
F	F	F	F	T	T	T

$(P \wedge Q) \Rightarrow R$ and $P \Rightarrow (Q \Rightarrow R)$ are equivalent according to the truth table

2. Negate the following: $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$

(Show proper steps) 4 points

$$\neg((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \quad \therefore \neg(P \Rightarrow Q) \vee \neg(Q \Rightarrow R)$$

$$\therefore \underline{\underline{(P \wedge \neg Q) \vee (Q \wedge \neg R)}}$$

3. Reduce the following augmented matrix to RREF and say if the solution set is unique/infinite or there are no solutions? **4 points**

(Explain why/why not)

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 3 & 6 & -3 & 9 \end{array} \right)$$

Aug. Form.

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 3 & 6 & -3 & 9 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 \end{array} \right]$$

$$R_2 = R_2 - 2R_1$$

$$R_3 = R_3 - 3R_1$$

The solution set is infinite as Rows 2 and 3 are empty rows.

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4. A fruit seller sells apples for \$2 each and oranges for \$3 each. One day, she sells 20 fruits for a total of \$52. How many apples and how many oranges did she sell?

6 points

Solve using Gauss Jordan elimination and show proper steps

$$\begin{aligned} a + o &= 20 \\ 2a + 3o &= 52 \end{aligned}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 20 \\ 2 & 3 & 52 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 20 \\ 0 & 1 & 12 \end{array} \right]$$

Aug. form.

$$R_2 = R_2 - 2R_1$$

$$\rightarrow \left[\begin{array}{cc|c} 1 & 0 & 8 \\ 0 & 1 & 12 \end{array} \right]$$

She sold 8 apples and 12 oranges.

$$R_1 = R_1 - R_2$$

5. Prove using mathematical induction that for all $n \geq 1$: 10 points

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{(n+1)}$$

(Write each step clearly)

Base condition :

$$f(n) = \frac{n}{n+1} \quad f(1) = \frac{1}{2} = \frac{1}{1+1} = \frac{1}{2} \rightarrow \text{base condition ok.}$$

inductive hypothesis: for some $k \geq 1$, $f(k+1)$ is true.

Solve

$$f(k+1) = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{(k+1) \cdot 1} + \frac{1}{(k+1)(k+2)}$$

$$\frac{k+1}{k+2} = \frac{k}{(k+1)} + \frac{1}{(k+1)(k+2)}$$

$$\frac{k+1}{k+2} = \frac{k^2 + 2k + 1}{(k+2)(k+1)}$$

$$k+1 = \frac{(k+1)^2}{k+2}$$

$$k+1 = k+1 \rightarrow \text{This is true.}$$

6. Given two vectors:

$$\vec{a} = (x, 2)$$

$$\vec{b} = (3, 1)$$

For what value of x will the two vectors be orthogonal/perpendicular? **5 points**

$$\vec{a} \cdot \vec{b} = x \cdot 3 + 2 \cdot 1 = 3x + 2$$

for a and b to be orthogonal, x has to be $-\frac{2}{3}$.



7. If A is a 2×3 matrix and B is a 3×4 matrix, what is the order of AB? What about BA? 4 points



AB =

$$\begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31}, A_{11}B_{12} + A_{12}B_{22} + A_{13}B_{32}, A_{11}B_{13} + A_{12}B_{23} + A_{13}B_{33}, A_{11}B_{14} + A_{12}B_{24} + A_{13}B_{34} \\ A_{21}B_{11} + A_{22}B_{21} + A_{23}B_{31}, A_{21}B_{12} + A_{22}B_{22} + A_{23}B_{32}, A_{21}B_{13} + A_{22}B_{23} + A_{23}B_{33}, A_{21}B_{14} + A_{22}B_{24} + A_{23}B_{34} \end{bmatrix}$$

BA is not a valid operations as the number of columns of B is not equal to the number of rows of A.

8. Let P be " $x^2 = 4$ " and Q be " $x = 2$ " 6 points

- Is P necessary or sufficient for Q?
- Write your answer as an implies statement

P is necessary for Q

$$P \Rightarrow Q$$

9. Given: "For every real number x , if $x > 1$ then $x^2 > 1$ ".

- Write the above statement using mathematical notations.
- Write the converse, Inverse and Contrapositive of the above statement. Use mathematical notations. **2 + 1 + 1 + 1 = 5 points**

$$\forall x \in \mathbb{R}, x > 1 \Rightarrow x^2 > 1.$$

$$\text{converse: } \forall x \in \mathbb{R}, x^2 > 1 \Rightarrow x > 1.$$

$$\text{inverse: } \forall x \in \mathbb{R}, x \leq 1 \Rightarrow x^2 \leq 1.$$

$$\text{contrapositive: } \forall x \in \mathbb{R}, x^2 \leq 1 \Rightarrow x \leq 1.$$

Extra Work -- Please clearly label which problem you are working on. Make sure to note on the original problem that your solution is continued here.